

ReadyCare MVP API Design

Updated version including reviews and professional registration document flow

This document defines the proposed MVP API for ReadyCare. It is designed to be clear enough to implement in Java Spring Boot, while still matching the main project requirements, flow diagrams, mock-ups, and draft data model.

1. MVP Scope and Design Approach

The API is based around three core user journeys: client, care professional, and admin. The essential MVP features are account creation, login, professional verification, availability management, booking, double-booking prevention, status updates, and reviews after completed care sessions.

Area	Included in MVP	Notes
Authentication	Yes	Client, professional and admin login. Client and professional registration are public.
Professional verification	Yes	Professionals upload documents after registration, then submit their application for admin review.
Availability	Yes	Approved professionals manage their availability calendar.
Booking	Yes	Clients search available professionals and create bookings.
Reviews	Yes	Clients and professionals can review each other after completed bookings.
Payments	Optional / later	Can be added later as a payment module without changing the core booking API.
Notifications	Basic / optional	Can be handled later through email or system notification records.

2. User Roles and Access Model

For implementation, keep the roles simple. More detailed admin permissions can be added later if needed.

Role	Purpose
CLIENT	Can manage their account, search professionals, create bookings, cancel own bookings, and leave reviews after completed sessions.
PROFESSIONAL	Can manage profile, upload verification documents, submit application, manage availability after approval, accept/reject bookings, and leave reviews after completed sessions.
ADMIN	Can view users, view applications, approve/reject professionals, view bookings, and manage user status.

3. Professional Registration and Document Upload Flow

The document upload and application submission endpoints should not be fully public. A public file upload endpoint would create security and storage risks, and it would be harder to link uploaded documents to a verified professional account. The safer design is to make professional registration public, return a JWT token after registration, and then use that token immediately during the same frontend flow to upload documents and submit the application.

Step	Endpoint	Auth Required?	Description
1	POST /api/auth/register/professional	No	Creates the professional account and returns an access token.
2	POST /api/professionals/me/documents	Yes - PROFESSIONAL	Uploads required documents such as DBS, right to work, care certificate, safeguarding, first aid or manual handling.
3	POST /api/professionals/me/submit-application	Yes - PROFESSIONAL	Submits the application for admin review and changes status to PENDING_APPROVAL.

This still feels like one registration flow to the professional, but the backend remains secure because uploads belong to an authenticated user.

```
{
  "userId": "uuid",
  "userType": "PROFESSIONAL",
  "certificationStatus": "NOT_STARTED",
  "accessToken": "jwt-token-here",
```

```

    "message": "Professional account created. Please upload your documents to complete your application."
  }
}

```

Certification Status	Meaning
NOT_STARTED	Professional has registered but has not submitted the full application yet.
PENDING_APPROVAL	Professional has uploaded documents and submitted the application for admin review.
APPROVED	Professional is approved and can appear in search results, manage bookable availability and receive bookings.
REJECTED	Professional application was rejected. They may be allowed to upload corrected documents.
SUSPENDED	Professional account has been disabled by admin.

4. Authentication Endpoints

Method	Path	Title	Description	Request Body	Response Body	Role
POST	/api/auth/register/client	Register Client	Creates a new client account and client profile.	firstName, lastName, email, password, phoneNumber, gender, dob, address	userId, userType, message	Public
POST	/api/auth/register/professional	Register Professional	Creates a professional account with certificationStatus NOT_STARTED and returns a JWT so the frontend can continue to document upload.	firstName, lastName, email, password, phoneNumber, gender, dob, address, bio, hourlyRateOfficeHours, hourlyRateOutOfOfficeHours	userId, userType, certificationStatus, accessToken, message	Public
POST	/api/auth/login	Login	Authenticates client, professional or admin and returns a JWT token.	email, password	accessToken, userId, userType, certificationStatus	Public
POST	/api/auth/logout	Logout	Optional endpoint if the frontend wants to call logout. JWT logout can also be handled client-side.	None	message	Authenticated

5. User and Profile Endpoints

Method	Path	Title	Description	Request Body	Response Body	Role
GET	/api/users/me	Get Current User	Returns the logged-in user profile.	None	User profile details	CLIENT, PROFESSIONAL, ADMIN
PATCH	/api/users/me	Update Current User	Updates basic details for the logged-in user.	firstName, lastName, phoneNumber, address	Updated user profile	CLIENT, PROFESSIONAL, ADMIN
PATCH	/api/professionals/me	Update Professional Profile	Allows professional to update bio and hourly rates.	bio, hourlyRateOfficeHours, hourlyRateOutOfOfficeHours	Updated professional profile	PROFESSIONAL
GET	/api/professionals/{professionalId}	Get Professional Details	Returns public profile details for the client booking flow.	None	professionalId, name, bio, hourlyRate, certificationStatus, averageRating	CLIENT, ADMIN or Public

6. Professional Document and Verification Endpoints

Method	Path	Title	Description	Request Body	Response Body	Role
POST	/api/professionals/me/documents	Upload Verification Document	Professional uploads required verification documents after registration.	documentType, file or fileUrl	documentId, documentType, status, message	PROFESSIONAL

			Does not require approval, but does require login.			
GET	/api/professionals/me/documents	View My Documents	Professional views uploaded documents and statuses.	None	List of uploaded documents	PROFESSIONAL
POST	/api/professionals/me/submit-application	Submit Application	Professional submits documents/profile for admin review.	None	certificationStatus: PENDING_APPROVAL, message	PROFESSIONAL
GET	/api/admin/professionals/pending	View Pending Professionals	Admin views professionals waiting for approval.	None	List of pending professionals	ADMIN
GET	/api/admin/professionals/{professionalId}	View Professional Application	Admin views professional profile and uploaded documents.	None	Professional application details	ADMIN
POST	/api/admin/professionals/{professionalId}/approve	Approve Professional	Admin approves professional application.	Optional: notes	professionalId, certificationStatus: APPROVED	ADMIN
POST	/api/admin/professionals/{professionalId}/reject	Reject Professional	Admin rejects professional application.	reason	professionalId, certificationStatus: REJECTED, reason	ADMIN

7. Availability Endpoints

Method	Path	Title	Description	Request Body / Query Params	Response Body	Role
POST	/api/availability	Create Availability Slot	Professional creates an available working slot.	startTs, endTs, status	availabilitySlotId, professionalId, startTs, endTs, status	PROFESSIONAL
GET	/api/availability/me	View My Availability	Professional views own availability slots.	Query: from, to	List of availability slots	PROFESSIONAL
PATCH	/api/availability/{slotId}	Update Availability Slot	Professional updates a slot.	startTs, endTs, status	Updated availability slot	PROFESSIONAL
DELETE	/api/availability/{slotId}	Delete Availability Slot	Professional deletes a slot if it has not been booked.	None	message	PROFESSIONAL
GET	/api/professionals/{professionalId}/availability	View Professional Availability	Client views a specific professional availability.	Query: from, to	List of available slots	CLIENT

Availability Validation Rule	Reason
Professional must be APPROVED before availability is visible to clients.	Prevents unverified carers being booked.
Slot start time must be before end time.	Basic time validation.
Slot should not overlap another slot for the same professional.	Keeps the calendar consistent.
Booked slots cannot be deleted directly.	Prevents breaking existing bookings.

8. Search and Booking Endpoints

Method	Path	Title	Description	Request Body / Query Params	Response Body	Role
GET	/api/professionals/search	Search Available Professionals	Finds approved professionals by city, date/time and duration.	Query: city, startTs, durationHours	List of available professionals	CLIENT
POST	/api/bookings	Create Booking	Client books a professional for	professionalId, addressId,	bookingId, status, price,	CLIENT

			a specific time slot.	startTs, endTs	message	
GET	/api/bookings/me	View My Bookings	Returns bookings for logged-in client or professional.	Query: status, from, to	List of bookings	CLIENT, PROFESSIONAL
GET	/api/bookings/{bookingId}	View Booking Details	Returns full details of one booking.	None	Booking details	Booking participant or ADMIN
POST	/api/bookings/{bookingId}/cancel	Cancel Booking	Client or professional cancels a booking.	reason	bookingId, status: CANCELLED, cancelledAt	CLIENT, PROFESSIONAL
POST	/api/bookings/{bookingId}/accept	Accept Booking	Professional accepts a booking request.	None	bookingId, status: CONFIRMED	PROFESSIONAL
POST	/api/bookings/{bookingId}/reject	Reject Booking	Professional rejects a booking request.	reason	bookingId, status: REJECTED	PROFESSIONAL
POST	/api/bookings/{bookingId}/complete	Complete Booking	Marks a booking as completed so reviews can be left. This could be triggered by admin or automatically after the session.	None or notes	bookingId, status: COMPLETED	ADMIN or system

Recommended booking statuses:

PENDING
 CONFIRMED
 CANCELLED
 REJECTED
 COMPLETED

9. Review Endpoints

Reviews are part of the MVP because they are important to building trust between clients and care professionals. A review should only be allowed after a booking has been completed. This prevents fake reviews from users who have not used the service.

Method	Path	Title	Description	Request Body / Query Params	Response Body	Role
POST	/api/bookings/{bookingId}/reviews	Leave Review	Allows a client or professional to review the other participant after a completed booking.	rating, comment	reviewId, bookingId, reviewerUserId, revieweeUserId, rating, comment, createdAt	CLIENT, PROFESSIONAL
GET	/api/professionals/{professionalId}/reviews	View Professional Reviews	Allows users to view reviews for a professional before booking.	None	List of reviews, averageRating, reviewCount	Public or CLIENT
GET	/api/users/me/reviews	View My Reviews	Allows logged-in clients/professionals to view reviews they have received.	None	List of received reviews	CLIENT, PROFESSIONAL
GET	/api/bookings/{bookingId}/reviews	View Booking Reviews	Shows reviews linked to a specific booking.	None	List of reviews for that booking	Booking participant or ADMIN
GET	/api/admin/reviews	View All Reviews	Allows admin to view and monitor reviews.	Query: rating, reviewerUserId, revieweeUserId	List of reviews	ADMIN

Review Validation Rule	Reason
Booking must be COMPLETED.	Prevents reviews before the care session has taken place.
Reviewer must be part of the booking.	Prevents unrelated users leaving reviews.
Reviewer can only review the other participant.	Client reviews professional; professional reviews client.
One review per reviewer per booking.	Prevents duplicate reviews.
Rating must be between 1 and 5.	Keeps review data consistent.

```

reviews
- id
- booking_id
- reviewer_user_id
- reviewee_user_id
- rating
- comment
- created_at

```

10. Admin Endpoints

Method	Path	Title	Description	Request Body / Query Params	Response Body	Role
GET	/api/admin/users	View All Users	Admin views all registered users.	Query: userType, status, search	List of users	ADMIN
GET	/api/admin/users/{userId}	View User Details	Admin views a specific user.	None	User details	ADMIN
PATCH	/api/admin/users/{userId}/status	Update User Status	Admin suspends or reactivates a user if needed.	status, reason	Updated user status	ADMIN
GET	/api/admin/bookings	View All Bookings	Admin views all platform bookings.	Query: status, from, to, professionalId, clientId	List of bookings	ADMIN
GET	/api/admin/bookings/{bookingId}	View Any Booking	Admin views full details of any booking.	None	Booking details	ADMIN

11. Example DTOs

Register Client Request

```

{
  "firstName": "Ameet",
  "lastName": "Bassan",
  "email": "ameet@example.com",
  "password": "Password123!",
  "phoneNumber": "07123456789",
  "gender": "MALE",
  "dob": "1999-01-01",
  "address": {
    "line1": "10 Example Road",
    "city": "London",
    "postcode": "SE9 5JX",
    "country": "GB"
  }
}

```

Register Professional Request

```

{
  "firstName": "Sarah",
  "lastName": "Mitchell",
  "email": "sarah@example.com",
  "password": "Password123!",
  "phoneNumber": "07123456789",
  "gender": "FEMALE",
  "dob": "1985-01-01",
  "bio": "Experienced care professional with elderly care experience.",
  "hourlyRateOfficeHours": 20.00,
  "hourlyRateOutOfOfficeHours": 28.00,
  "address": {
    "line1": "20 Care Street",
    "city": "London",

```

```

    "postcode": "SE10 1AA",
    "country": "GB"
  }
}

```

Upload Document Request

```

POST /api/professionals/me/documents
Content-Type: multipart/form-data

```

```

documentType = DBS_CERTIFICATE
file = dbs-certificate.pdf

```

Create Booking Request

```

{
  "professionalId": "uuid",
  "addressId": "uuid",
  "startTs": "2026-05-01T10:00:00",
  "endTs": "2026-05-01T12:00:00"
}

```

Create Review Request

```

{
  "rating": 5,
  "comment": "The care professional was punctual, friendly and helpful."
}

```

12. Status Codes

Status Code	Meaning	Example
200 OK	Request succeeded	Login, get profile, view bookings
201 Created	New resource created	Register user, create booking, create availability, create review
400 Bad Request	Invalid input	End time before start time, rating outside 1-5
401 Unauthorized	Not logged in	Missing or invalid JWT
403 Forbidden	Logged in but not allowed	Client tries to approve professional
404 Not Found	Resource does not exist	Booking ID not found
409 Conflict	Resource conflict	Double booking attempt or duplicate review for same booking
422 Unprocessable Entity	Valid JSON but business rule failed	Professional not approved, booking not completed so review cannot be created
500 Internal Server Error	Unexpected backend error	Database failure

13. Key Business Rules

Area	Rule
Professional approval	Only APPROVED professionals can appear in search results and receive bookings.
Document upload	Documents can be uploaded during the registration journey, but only after the professional account is created and the user has a JWT token.
Booking availability	Bookings must fit within a FREE availability slot.
Double-booking prevention	A professional cannot have overlapping PENDING or CONFIRMED bookings.
Booking time rules	For MVP, bookings should start and finish on the hour.
Reviews	Reviews can only be created after a booking is COMPLETED.
Review ownership	Only booking participants can leave or view booking-specific reviews.
Admin approval	Admin controls whether a professional becomes APPROVED, REJECTED or SUSPENDED.

Recommended overlap check:

```
existing.start_ts < requested.end_ts
AND existing.end_ts > requested.start_ts
AND existing.status IN ('PENDING', 'CONFIRMED')
```

14. Final Recommended MVP Endpoint List

```
/api/auth/register/client
/api/auth/register/professional
/api/auth/login

/api/users/me
/api/professionals/me
/api/professionals/me/documents
/api/professionals/me/submit-application
/api/professionals/search
/api/professionals/{professionalId}
/api/professionals/{professionalId}/reviews

/api/availability
/api/availability/me
/api/availability/{slotId}

/api/bookings
/api/bookings/me
/api/bookings/{bookingId}
/api/bookings/{bookingId}/cancel
/api/bookings/{bookingId}/accept
/api/bookings/{bookingId}/reject
/api/bookings/{bookingId}/complete
/api/bookings/{bookingId}/reviews

/api/users/me/reviews
/api/admin/reviews

/api/admin/users
/api/admin/users/{userId}
/api/admin/professionals/pending
/api/admin/professionals/{professionalId}
/api/admin/professionals/{professionalId}/approve
/api/admin/professionals/{professionalId}/reject
/api/admin/bookings
/api/admin/bookings/{bookingId}
```